

Parallel Robots

EPFL – 2026

Arda YIĞIT



Singularity Analysis

- Consider a parallel robot such that

$$\mathbf{A}(\mathbf{x}, \mathbf{q})\dot{\mathbf{x}} + \mathbf{B}(\mathbf{x}, \mathbf{q})\dot{\mathbf{q}} = \mathbf{0}$$

where $\mathbf{A}$ and $\mathbf{B}$ are square matrices

- Definition: a configuration is singular if either $\mathbf{A}$ or $\mathbf{B}$ is singular
- With this definition, a configuration is singular for a given set of $\mathbf{q}$ and $\mathbf{x}$

- This is the historical definition used by Gosselin (1990)
- More recently, more complete definitions were proposed, but are out of scope
- One of the most important and difficult topic for the study of parallel robot
- Parallel robots exhibit different singularities compared to serial architectures

- Definition: the first kind of singularity occurs when

$$\det \mathbf{B} = 0$$

- Equivalently: there exist $\dot{\mathbf{q}} \neq \mathbf{0}$ such that $\dot{\mathbf{x}} = \mathbf{0}$, so $\dot{\mathbf{q}} \in \ker \mathbf{B}$
- Same as the singularity of a serial robot

- Different branches (solutions) of the IK meet
- Physical interpretation
 - Workspace boundary
 - The robot loses the ability to move along one direction
 - The actuators require infinite force to move along one direction

- Definition: the second kind of singularity occurs when

$$\det \mathbf{A} = 0$$

- Equivalently: there exist $\dot{\mathbf{x}} \neq \mathbf{0}$ such that $\dot{\mathbf{q}} = \mathbf{0}$, so $\dot{\mathbf{x}} \in \ker \mathbf{A}$
- Only for parallel robots
- Happens within the workspace

- Different branches (solutions) of the FK meet
- Physical interpretation
 - The end-effector can move even when all actuators are locked
 - The actuators require infinite force to resist to some disturbances

- Identify serial and parallel singularities of 5-bar and 3-RRR robots
 - Geometrical interpretation

Serial singularities

- Most of the time, it is enough to analyze $\det \mathbf{B}$
- Often, $\mathbf{B}$ is block diagonal
- Can be geometrically “intuitive” as it describes the boundaries of the workspace
- Often, it is sufficient to analyze the leg mechanism

Parallel singularities

- In most cases, there are no closed-form solutions to $\det \mathbf{A} = 0$
- Geometric interpretation in necessary
- Numerical solutions do not help geometric understanding

- Let us revisit the differential kinematics obtained with screw theory approach

$$\begin{bmatrix} \zeta_1^T \\ \zeta_2^T \\ \dots \\ \zeta_n^T \end{bmatrix} \mathbf{t}_p = \begin{bmatrix} \zeta_1^T \$_{1a} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \zeta_2^T \$_{2a} & \dots & \mathbf{0} \\ \dots & \dots & \dots & \dots \\ \mathbf{0} & \mathbf{0} & \dots & \zeta_n^T \$_{na} \end{bmatrix} \dot{\mathbf{q}}$$

- Under some assumptions... (which ones?)
- Serial singularity if $\zeta_i^T \$_{ia} = 0$
 - The screw that is reciprocal to passive joints is also reciprocal to the active one(s)
- Parallel singularity if there is a linear dependency between $(\zeta_i)_i$

- Serial singularity example – $\underline{R}RR$ and $R\underline{P}R$ legs

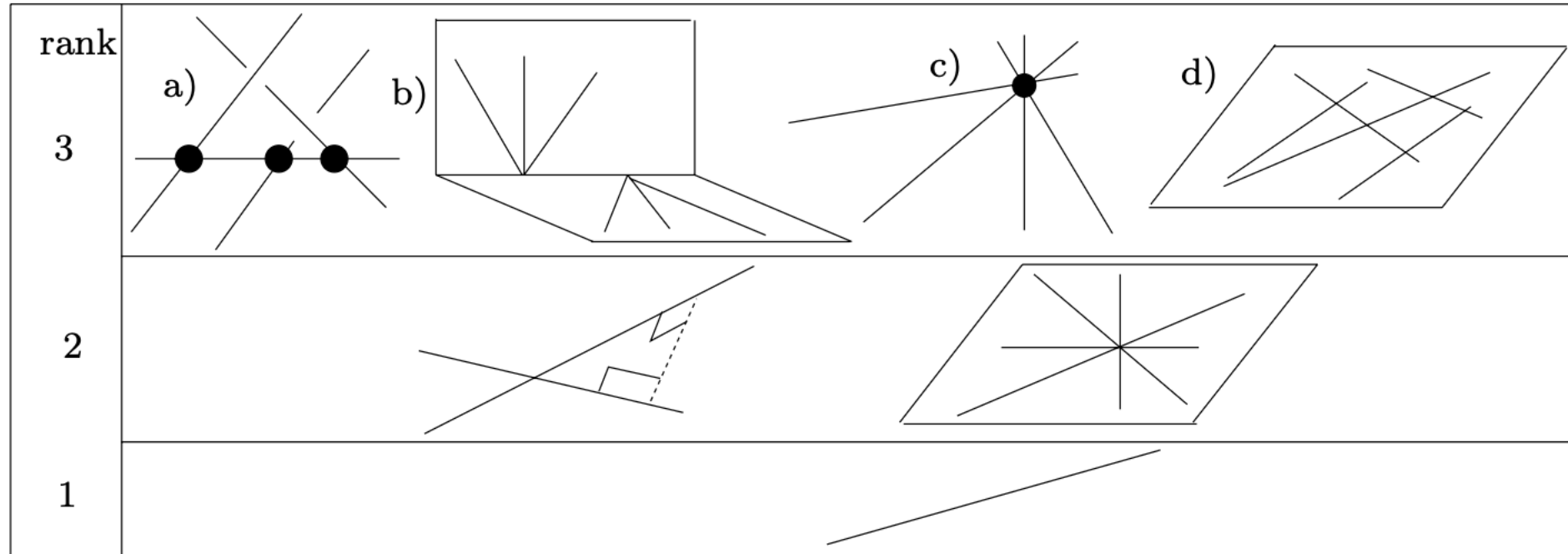
- Parallel singularity if there is a linear dependency between $(\zeta_i)_i$
- How to detect geometrically a linear dependency?
 - Grassmann geometry
 - Grassman-Cayley algebra

- Definition: the Plücker coordinates of a line of direction $\mathbf{f}$ passing through a point P are

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{r}_{OP} \times \mathbf{f} \end{bmatrix}$$

- Theorem: for any spatial (planar) robot whose legs are attached to the end-effector through a spherical (revolute) joint, $\mathbf{A}$ matrix will involve the Plücker vector of a line associated to the link connecting the leg to the end-effector
- In layman's terms, ζ_i is a Plücker vector

- Grassmann geometry allows for evaluating the rank of the A matrix



<https://doi.org/10.1007/1-4020-4133-0>

- Grassmann geometry allows for evaluating the rank of the A matrix
- For 2-DOF planar robots, rank deficiency if all two lines are coincident
- For 3-DOF planar robots, rank deficiency if all three lines are concurrent
 - Parallel lines are concurrent at infinity
- 3D case is out-of-scope